AWS Certified Data Engineer Associate: Architecture Scenarios

A Visual Study Guide for Scenarios

3

Unofficial Study Guide based on 'sthithapragna' Exam Prep

NotebookLM

Optimizing AWS Glue for Small Files

Challenge:

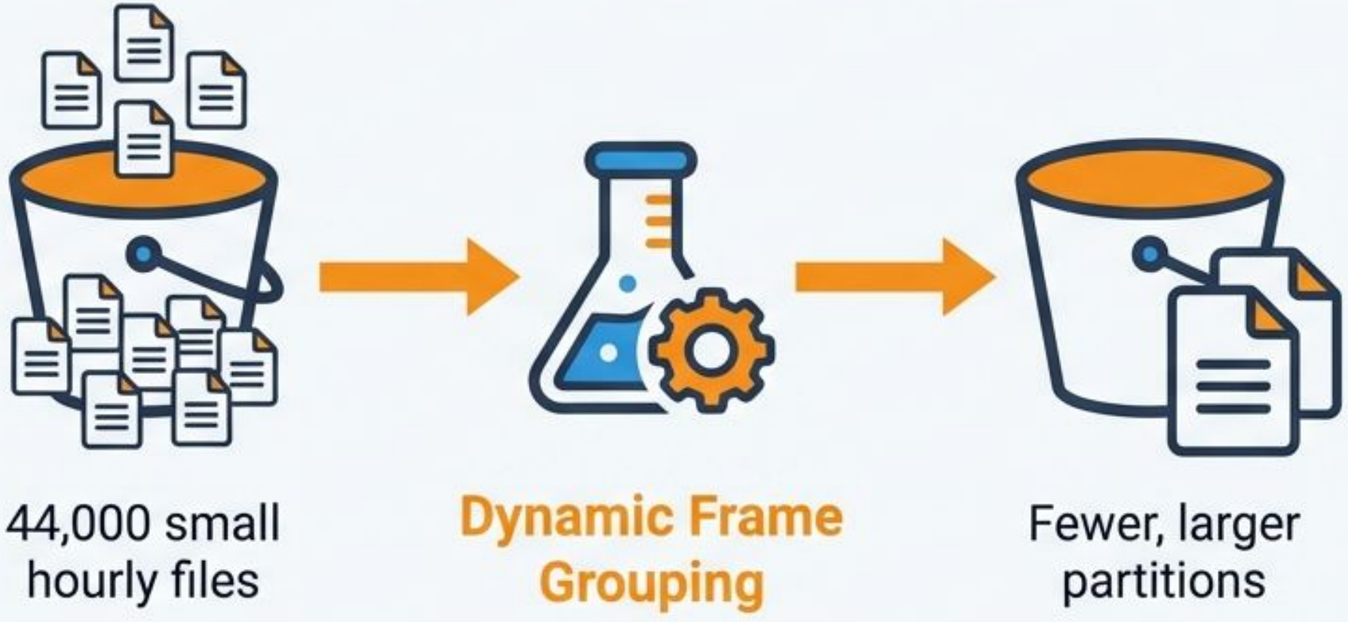
Processing tens of thousands of small, hourly files (200-300 KB) in S3 causes extremely poor Glue pipeline performance due to massive metadata overhead.

Solution:

Enable Dynamic Frame Grouping in AWS Glue to automatically merge small files into larger partitions before processing.

Exam Key:

This feature significantly reduces read/write operations and improves Apache Spark parallelization without requiring larger, more expensive worker nodes.



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Integrating SaaS Applications for Analytics

Challenge:

Extract, combine, and analyze data from multiple third-party SaaS applications using QuickSight, with minimal custom development.

Solution:

Use Amazon AppFlow to securely fetch and ingest SaaS data directly into AWS data stores.

Exam Key:

AppFlow is the designated fully managed, no-code integration service for connecting external SaaS platforms to the AWS analytics suite.

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Real-Time Log Analytics

Challenge:
Query and visualize server or application logs in near real-time with minimal delay to identify points of failure.

Solution:
Stream log data continuously using Amazon CloudWatch Logs subscription filters routed directly to Amazon OpenSearch Service.

Exam Key:
OpenSearch provides the necessary indexing and dashboarding capabilities for near real-time operational log analytics.

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Optimizing S3 Storage for Analytics

Challenge:

An e-commerce pipeline stores millions of daily orders as CSV files in S3. Analysts query this via Athena, causing high storage and data-retrieval costs.

Solution:

Convert the CSV output to Parquet format using AWS Glue ETL.

Exam Key:

Parquet is a columnar storage format. When paired with Snappy compression, it drastically reduces the S3 storage footprint and limits the data Athena needs to scan, lowering query costs.

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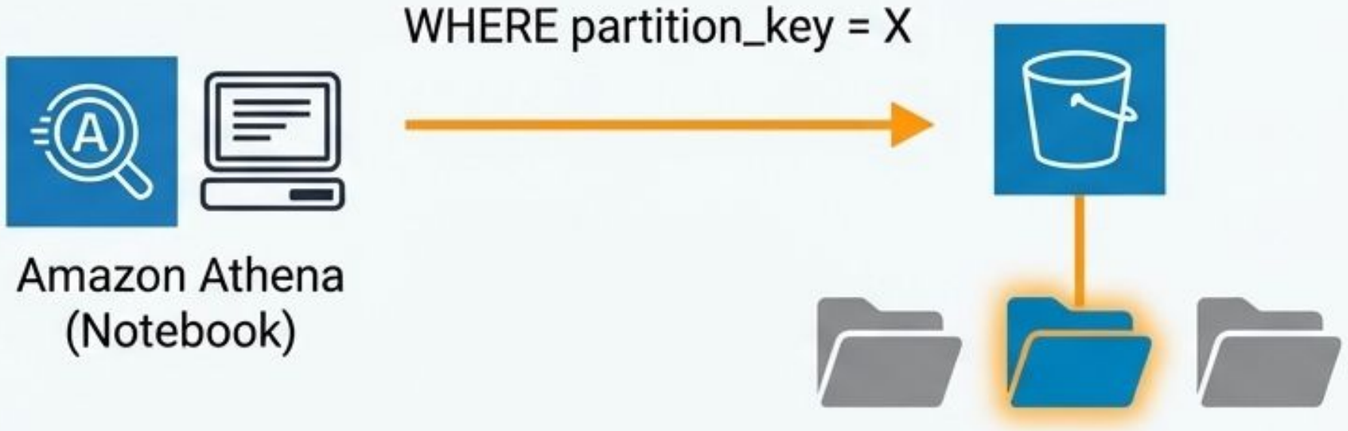
Maximizing Athena Query Efficiency

Challenge: Athena notebooks using Apache Spark are scanning too much data when querying partitioned S3 data, driving up costs.

Solution: Enforce Explicit Partition Pruning in the query logic.

Exam Key: Partition pruning only occurs if query filters explicitly reference the partition keys. This forces Athena to skip irrelevant S3 irrelevant S3 partitions, even if the Glue Catalog is fully up to date.



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Cross-Account QuickSight to Redshift Connectivity

Challenge:
QuickSight (Account A) must query an Amazon Redshift cluster located in a public subnet (Account B) to refresh dashboards.

Solution: Modify the Redshift Cluster Security Group in Account B to explicitly allow inbound traffic from the QuickSight resources.

Exam Key: Since the subnet is public, complex peering is unnecessary. Simply open the Redshift port (5439) to the QuickSight security group. QuickSight security group.

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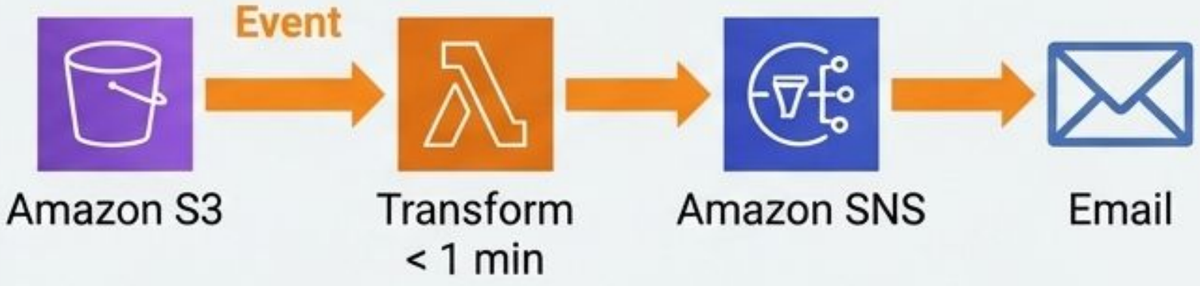
Lightweight S3 Event Processing

Challenge: Perform a sub-1-minute transformation whenever a new file is uploaded to S3 (approx. 1 file/hour), and email the result.

Solution: Use S3 Event Notifications to directly trigger an AWS Lambda function. Lambda transforms the data and publishes to an Amazon SNS topic.



Exam Key: For transformations taking under a minute, AWS Lambda is far more cost-effective and responsive than spinning up AWS Glue or EMR clusters.

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Handling Invalid Data in PySpark

Challenge: A Glue PySpark job fails repeatedly because the raw S3 data contains inconsistent values (e.g., special characters in a postal code).

Solution: Explicitly define the PySpark Schema within the Glue job script.

Exam Key: By defining the schema in code, the job can enforce data types, coerce formats, and safely reject or handle invalid records before the heavy transformation phase begins.

Define the schema explicitly and set mode=DROPMALFORMED to silently discard invalid records instead of failing the job

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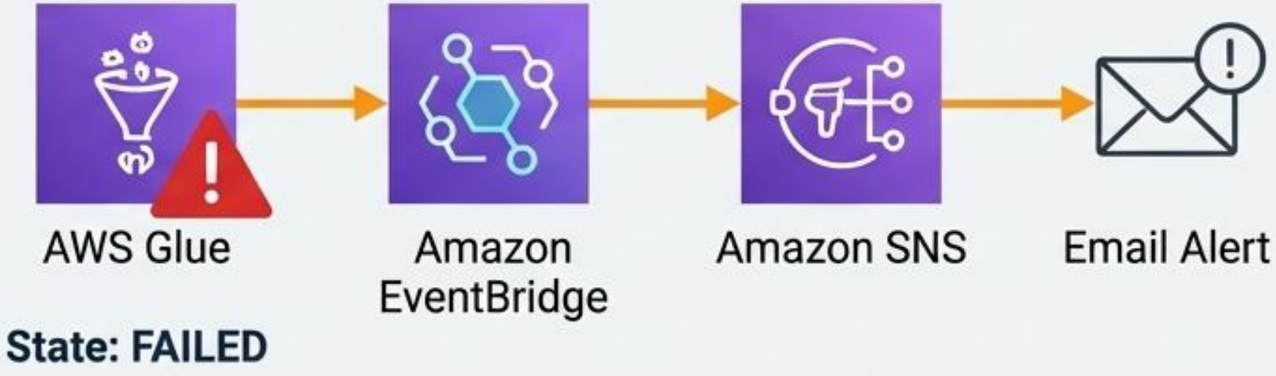
Automating Data Quality Alerts

Challenge: A Glue workflow fails due to data quality constraints. The reporting team requires immediate, automated email alerts.

Solution:
Configure an Amazon EventBridge rule to capture the AWS Glue job state-change events (e.g., FAILED).

Exam Key: EventBridge seamlessly catches Glue lifecycle events and routes them to an Amazon SNS topic subscribed to by the business team.



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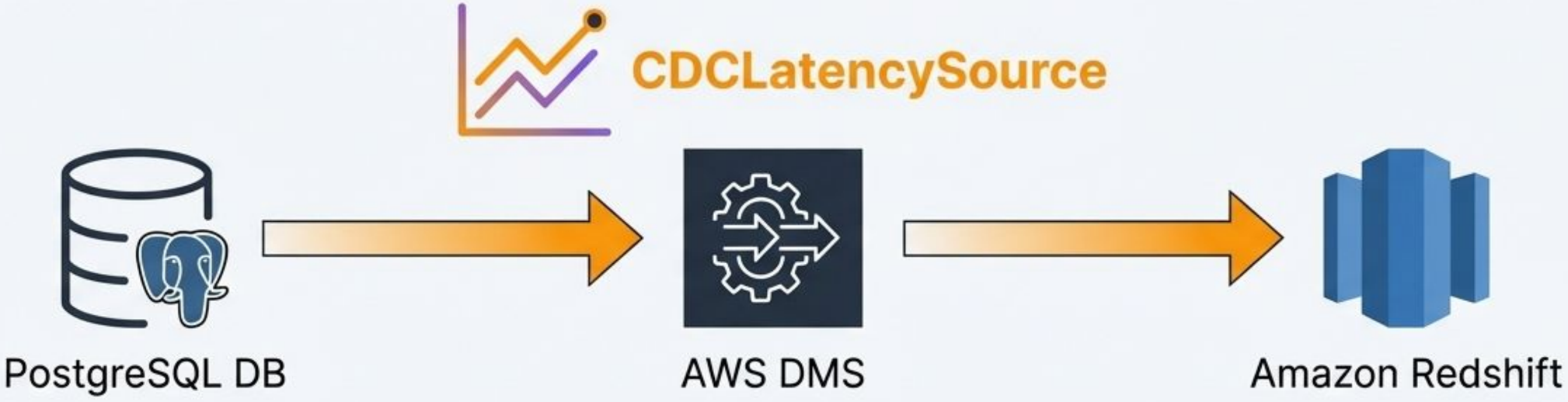
Isolating CDC Latency in Database Replication

Challenge: AWS DMS replication from PostgreSQL to Redshift shows high CDC latency. Need to confirm if the source database is the bottleneck.



Key Takeaways:

- Monitor CDCLatencySource to isolate delays occurring between the commit and the DMS read.
- High values directly indicate source-side lag, typically stemming from WAL or I/O constraints.
- Always differentiate latency metrics from throughput metrics during troubleshooting.

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Achieving Low-Latency IoT Data Ingestion

Challenge: IoT sensors send 100 KB every 10 seconds; data must reach S3 within a strict 30-second total latency window.

Key Takeaways:

- Streams combined with KCL consumers offer fine-grained flush control necessary for low-latency writes.
- Firehose imposes a minimum buffer limit of approximately 60 seconds, rendering it too slow for this case.
- KCL buffering allows data to reach S3 well within the 30-second requirement.

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Orchestrating Large-Scale ETL and Historical Processing

Challenge: Need daily 300 GB ETL and one-time TB-scale historical transformation on S3, all fully orchestrated.

Key Takeaways:
- Glue Crawlers automate schema inference as new data lands.
- EMR with Apache Spark handles both large daily ETL and massive batch historical workloads efficiently.
- MWAA ties both systems together into a unified, repeatable DAG workflow.

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Enforcing Business-Specific Data Validation

Challenge: Glue ETL on customer orders must enforce specific, business-defined data validation rules prior to processing.

Key Takeaways:

- Rulesets natively define specific checks like nulls, expected ranges, and uniqueness.
- Data Quality rules integrate directly into Glue ETL scripts, enabling inline pass/fail logic.
- The service automatically emits metrics to CloudWatch for dashboarding and operational alerts.

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Cost-Effective Querying of Compressed S3 Data

Challenge: Insurance company stores gzip-compressed transaction data and requires occasional, low-cost querying capabilities.

Key Takeaways:

- Amazon Athena reads gzip-compressed CSV and JSON files directly without requiring manual extraction.
- The serverless, pay-per-scan model makes it the ideal financial fit for infrequent audits.
- While gzip is an unsplittable format, it performs sufficiently well for occasional analytical workloads.

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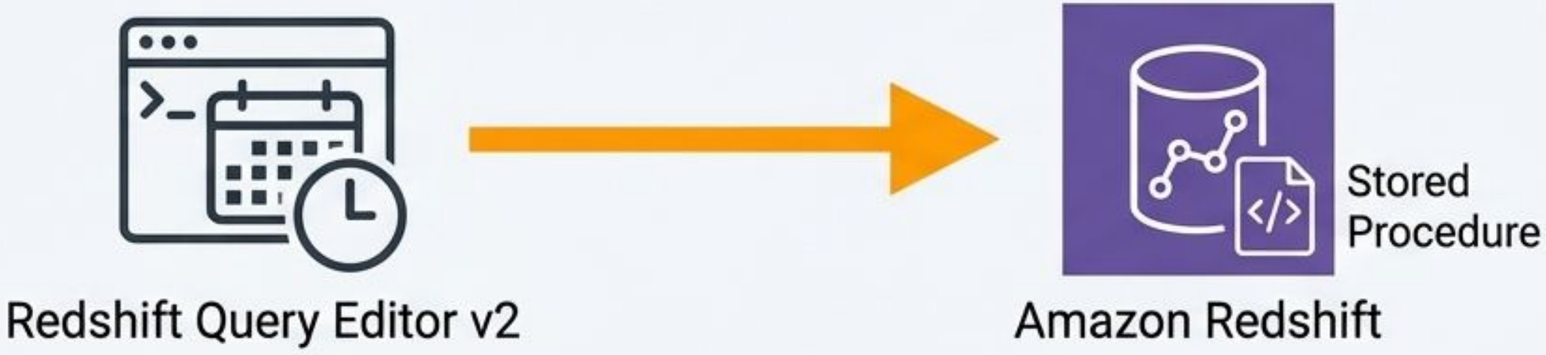
Automating Redshift Stored Procedures

Challenge: A Redshift stored procedure must run automatically each day with minimal setup and operational overhead.



Key Takeaways:
- Utilize Redshift Query Editor v2's built-in scheduling capabilities directly.
- Avoids the architectural complexity of adding external tools like AWS Lambda or AWS Glue.
- Keeps execution and monitoring centralized within Redshift, ensuring low ops overhead for simple daily jobs.

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Scaling Cost-Effective Daily Dashboards

Challenge: Clickstream data lands in S3 via Firehose; need daily QuickSight dashboards that scale rapidly and cost-effectively.

Key Takeaways:

- Athena provides serverless, pay-per- query SQL execution directly against S3 data lakes.
- QuickSight's SPICE engine ingests the results, enabling high-concurrency, cached dashboards without incurring further query costs.
- Utilizing a scheduled daily refresh perfectly balances data freshness requirements with budget constraints.

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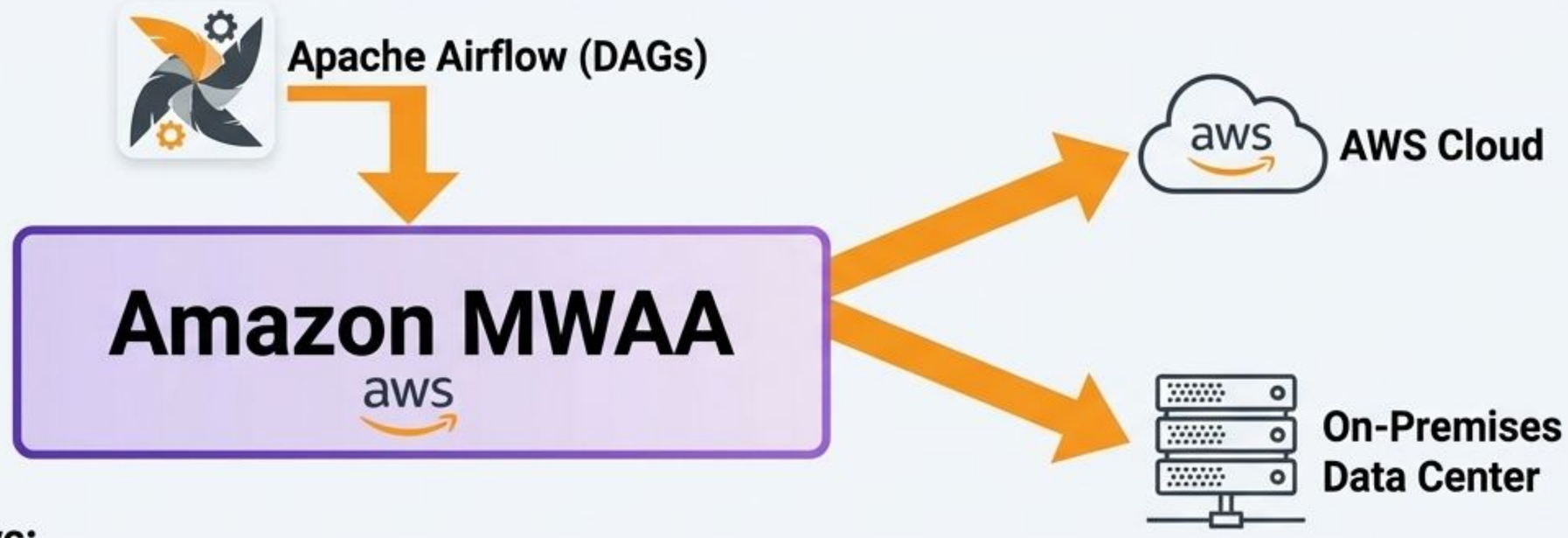
Implementing Portable, Open-Source Orchestration

Challenge: A data engineer needs portable pipeline orchestration across both on-premises and AWS environments utilizing open-source technology.



Key Takeaways:

- Standard Airflow DAGs remain fully portable across on-premises servers and the AWS cloud.
- MWAA handles all underlying infrastructure provisioning, security, and dynamic scaling automatically.
- Ensures consistent orchestration methodology utilizing a universally recognized open-source standard.

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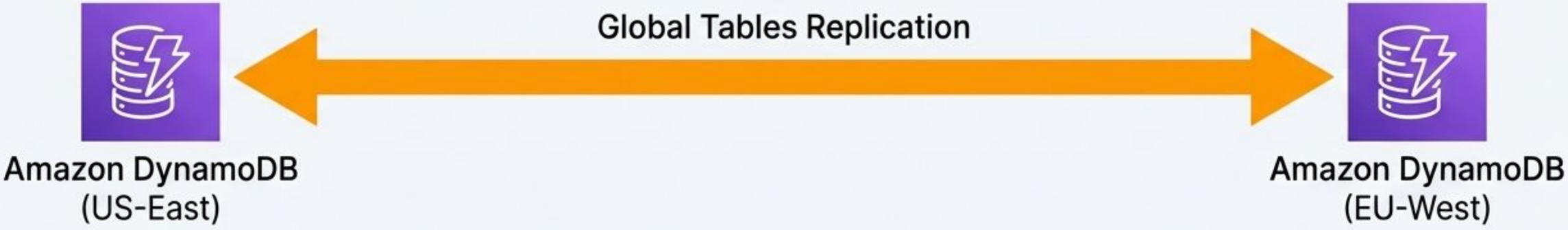
Deploying Globally Distributed, Low-Latency NoSQL

Challenge: A gaming company requires a globally distributed NoSQL database delivering sub-10 ms latency for high-volume OLTP workloads.



* Eventually consistent reads

Key Takeaways:

- DynamoDB acts as a fully managed, serverless NoSQL datastore requiring zero infrastructure provisioning.
- It guarantees consistent, single-digit millisecond latency for reads and writes at any scale.
- Global Tables provide active-active, multi-region replication automatically handled by AWS.

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Resolving Event-Driven Invocation Permissions

Challenge:

An EventBridge rule designed to invoke an AWS Lambda function fails repeatedly with an AccessDeniedException.

Key Takeaways:

- A Lambda function's resource policy explicitly controls which external services are permitted to invoke it.
- EventBridge must be granted specific lambda:InvokeFunction rights within that policy.
- Access denials occur when the invoking role and the target resource policy do not align; both must permit the action.

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Meeting Compliance with Dual-Layer S3 Encryption

Challenge: A company must comply with strict dual-layer encryption requirements for S3 data, utilizing only AWS-native capabilities.

DSSE-KMS

Key Takeaways:

- Dual-layer server-side encryption (DSSE-KMS) automatically applies two independent, KMS-managed encryption layers.
- It seamlessly satisfies rigorous regulatory compliance mandates requiring multi-layer encryption at rest.
- The entire process is fully managed within S3, completely eliminating the need for custom Lambda functions or complex client-side logic.

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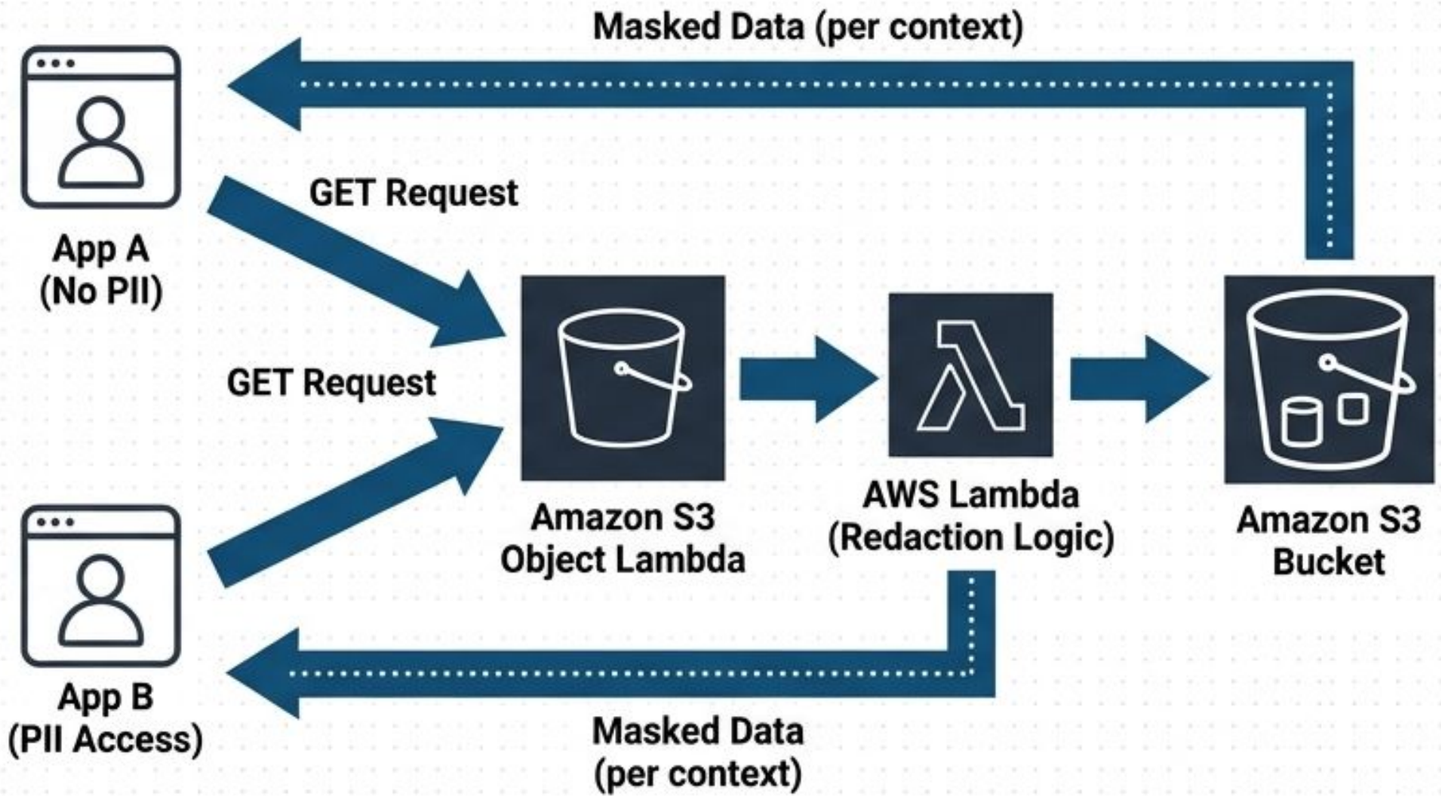
Dynamic PII Redaction in Amazon S3

Challenge:

Multiple applications (some requiring PII, others restricted) share identical datasets in S3. There is a critical need to **redact PII dynamically without duplicating** the underlying data.

Brief Explanation:

- S3 **Object Lambda** applies logic directly on standard GET requests.
- Maintains a single S3 data source, completely eliminating duplication and preprocessing.
- Dynamic masking adapts instantly based on the specific app or IAM context.
- Fully serverless, pay-per-request, and highly compliance-friendly.



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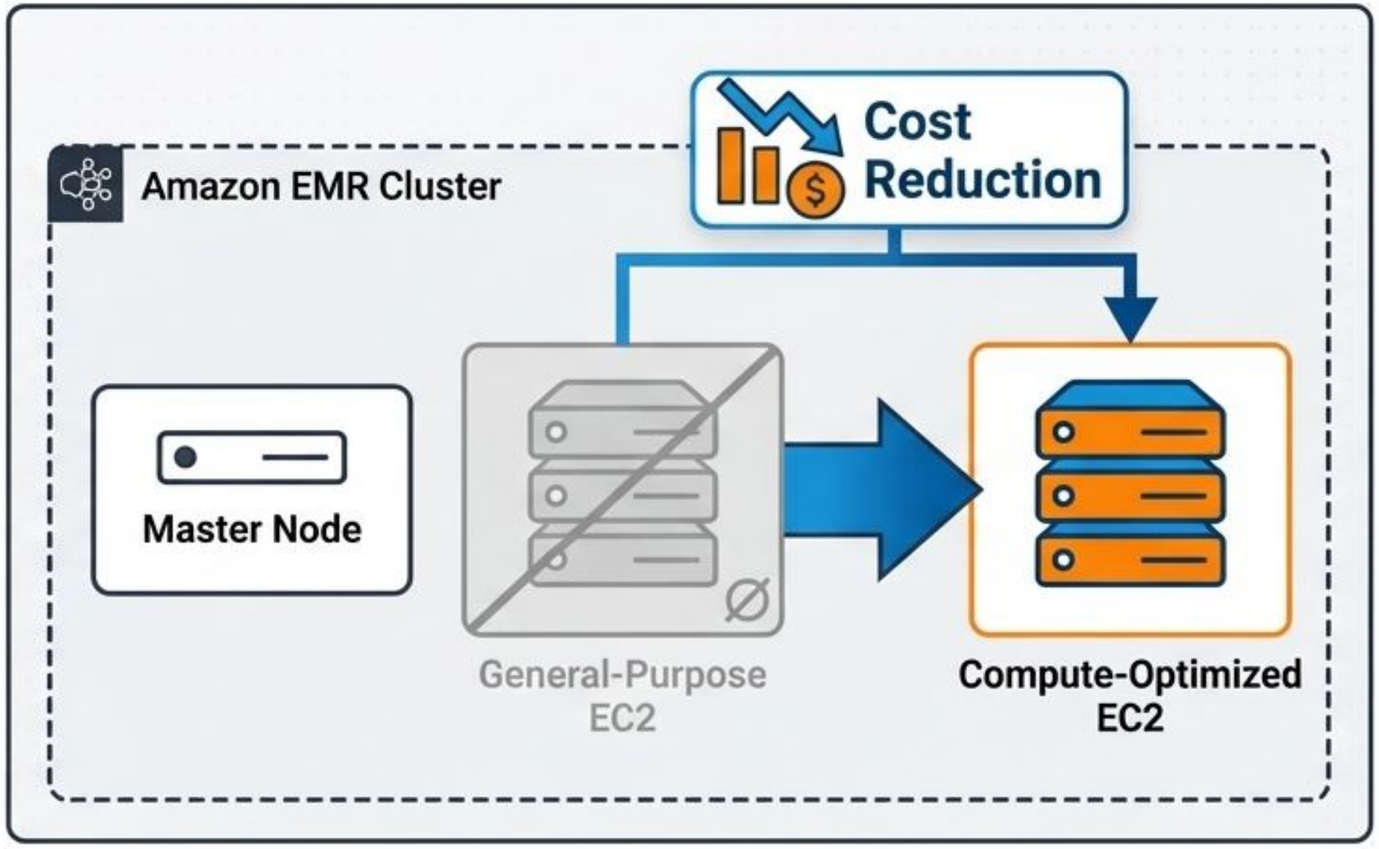
Cost Optimization for CPU-Bound Spark ETL

Challenge:

A Spark ETL job running on Amazon EMR is heavily **CPU-bound** but exhibits very **low memory usage**. The organization needs to **reduce infrastructure costs** without sacrificing pipeline performance.

Brief Explanation:

- Switch task nodes from general-purpose to compute-optimized EC2 instances.
- Compute-optimized EC2 types are the ideal fit for CPU-heavy Spark jobs.
- A consistently low memory footprint is the primary indicator of a compute-bound workload.
- Directly reduces EMR operational costs while maintaining necessary processing power.



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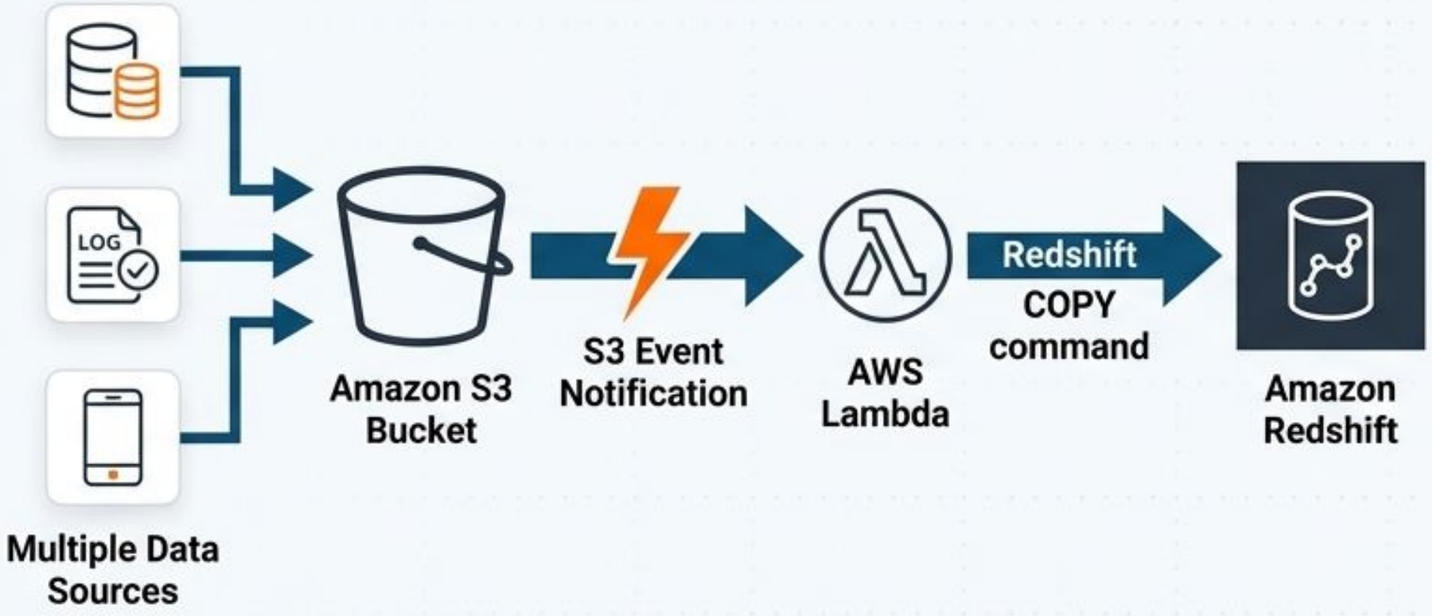
Near Real-Time S3 to Redshift Ingestion

Challenge:

Files from multiple external data sources are regularly uploaded to an S3 bucket. The data must be loaded into Amazon Redshift **as soon as they arrive** to achieve **near real-time** ingestion.

Brief Explanation:

- S3 Event Notifications provide instant, real-time triggers for every new file upload.
- AWS Lambda automates the ingestion process by invoking the highly efficient Redshift COPY command on each file.
- Achieves near real-time ETL entirely without manual scheduling or batch delays.



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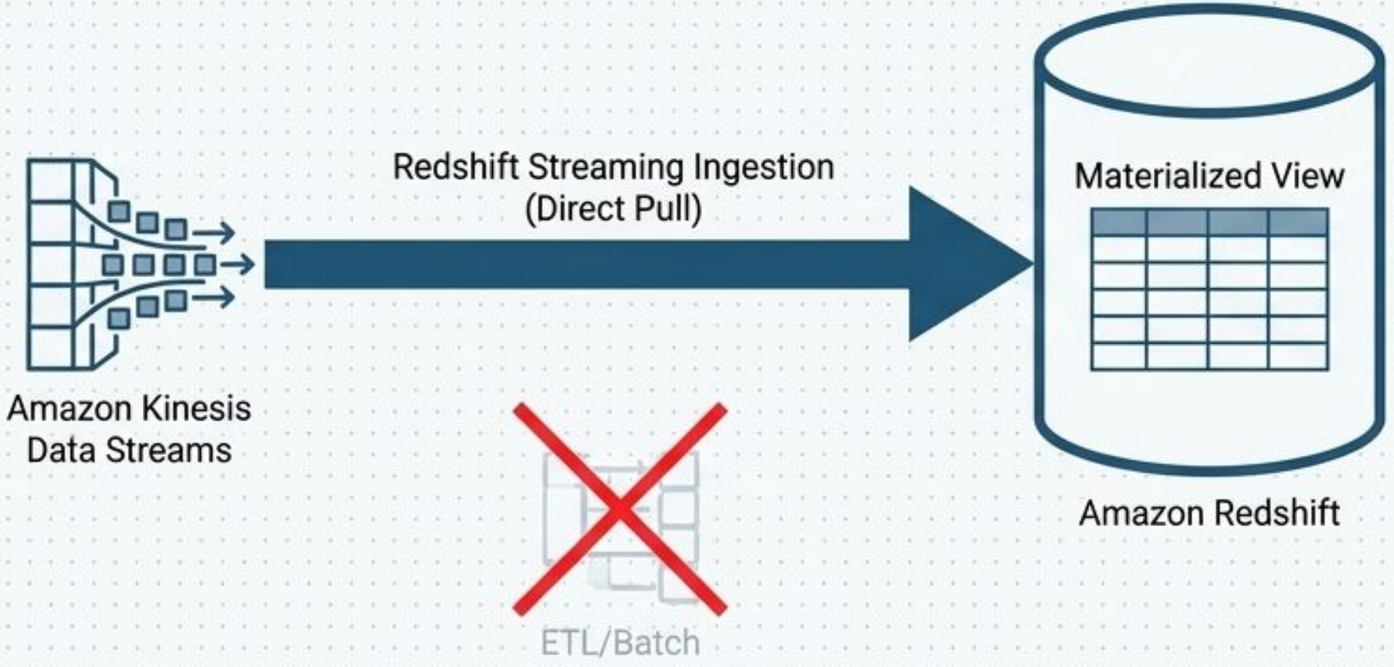
Low-Latency Kinesis to Redshift Streaming

Challenge:

A system uses Amazon Kinesis Data Streams for real-time log collection. This data must be ingested into Amazon Redshift for **immediate querying** and enrichment with the absolute **minimal management effort**.

Brief Explanation:

- Redshift streaming ingestion pulls data directly from Kinesis Data Streams, bypassing staging layers.
- Provides direct, low-latency integration.
- Data is immediately available in materialized views for downstream querying.
- Zero intermediate ETL or batch processing required, resulting in the lowest possible operational overhead.



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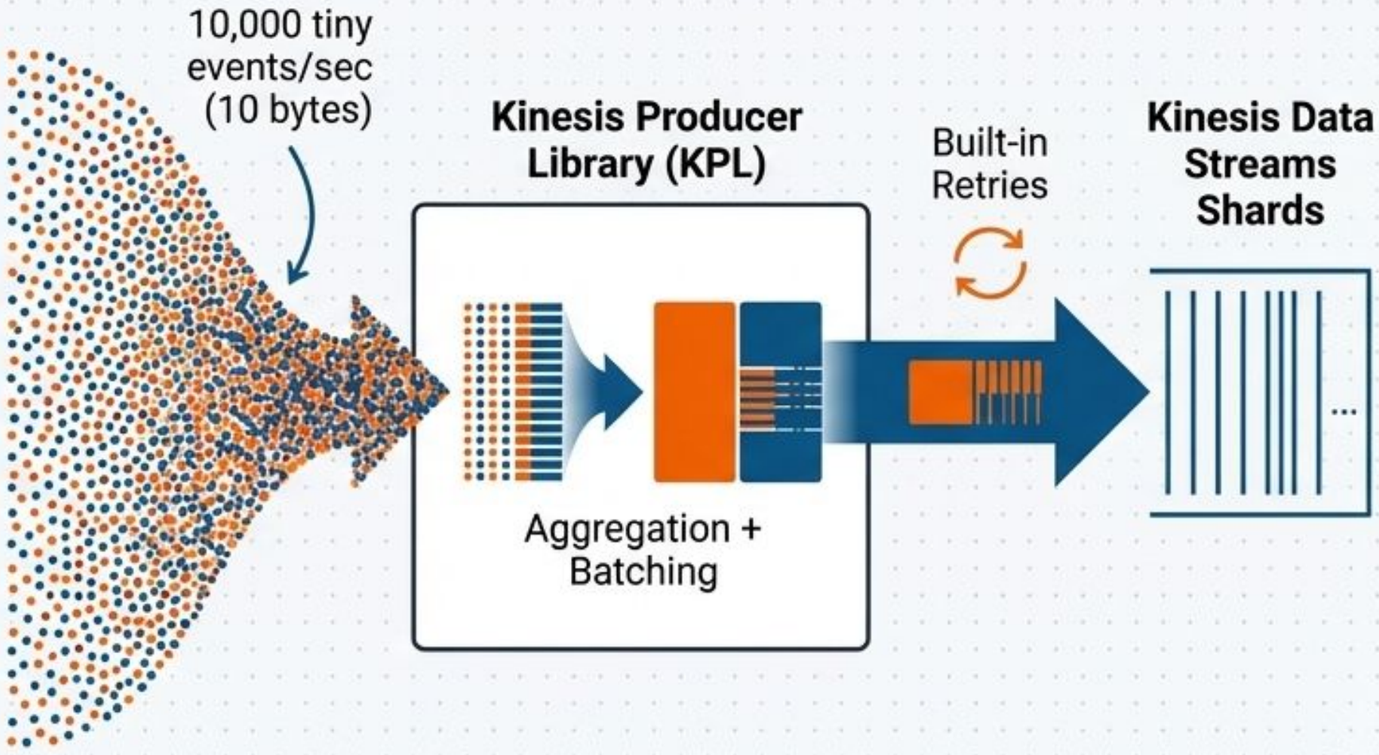
High-Throughput Micro-Event Ingestion into Kinesis

Challenge:

A system must reliably ingest up to **10,000 tiny geolocation events per second** (10 bytes each) into Kinesis Data Streams. The goal is to **maximize shard throughput** efficiency with **minimal ops**, where a few minutes of delay is acceptable.

Brief Explanation:

- Utilize the Kinesis Producer Library (KPL) instead of writing custom SDK code.
- KPL's built-in aggregation and batching maximizes shard efficiency specifically for tiny events.
- Native retries and buffering automatically improve ingestion reliability.
- Significantly lowers operational complexity compared to maintaining custom producer applications.



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